Current	Result	Size	Time	Cycles	GPU	SM Frequency	Process	Attributes
1673 - circuit_kernel_batch_plain_stack	(98, 1, 1)x(128, 1, 1)	6.35 ms	9,583,019	0 - NVIDIA GeForce RTX 2060 SUPER	1.47 GHz		[7728] python3.11	

Summary | Details | Source | Context | Comments | Raw | Session

Compare | Tools | View | Export

GPU Speed Of Light Throughput

GPU Throughput Chart

High-level overview of the throughput for compute and memory resources of the GPU. For each unit, the throughput reports the achieved percentage of utilization with respect to the theoretical maximum. Breakdowns show the throughput for each individual sub-metric. Compute and Memory to clearly identify the highest contributor. High-level overview of the utilization for compute and memory resources of the GPU presented as a roofline chart.

Compute (SM) Throughput [%]	35.13
Memory Throughput [%]	11.72
L1/TEX Cache Throughput [%]	13.47
L2 Cache Throughput [%]	6.79
DRAM Throughput [%]	11.72

Duration [ms]	6.35
Elapsed Cycles [cycle]	9588019
SM Active Cycles [cycle]	9075575.32
SM Frequency [GHz]	1.47
DRAM Frequency [GHz]	6.81

OK **Small Grid** This kernel grid is too small to fill the available resources on this device, resulting in only 0.36 full waves across all SMs. Look at Launch Statistics for more details.

Performance Indicators

Roofline Analysis The ratio of peak float (FP32) to double (FP64) performance on this device is 32:1. The workload achieved close to 0% of this device's FP32 peak performance and 11% of its FP64 peak performance. See the Profiling Guide for more details on roofline analysis.

GPU Throughput

PM Sampling

Timeline view of PM metrics sampled periodically over the workload duration. Data is collected across multiple passes. Use this section to understand how workload behavior changes over its runtime.

Maximum Sampling Interval [cycle]	20000	# Pass Groups 1
Maximum Buffer Size [Mbyte]	2.03	-

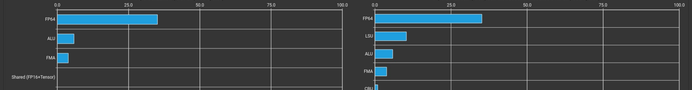
Compute Workload Analysis

Pipe Utilization (Elapsed Cycles)

Detailed analysis of the compute resources of the streaming multiprocessors (SM), including the achieved instructions per clock (IPC) and the utilization of each available pipeline. Pipelines with very high utilization might limit the overall performance.

Executed Ipc Elapsed [inst/cycle]	0.34	SM Busy [%] 35.13
Executed Ipc Active [inst/cycle]	0.35	Issue Slots Busy [%] 8.68
Issued Ipc Active [inst/cycle]	0.36	

Balanced FP64 is the highest-utilized pipeline (35.1%) based on elapsed cycles in the workload, taking into account the rates of its different instructions. It executes 64-bit floating point operations. It is well-utilized, but should not be a bottleneck.



Memory Workload Analysis

Memory Chart

Detailed analysis of the memory resources of the GPU. Memory can become a limiting factor for the overall kernel performance when fully utilizing the involved hardware units (Mem Busy), exhausting the available communication bandwidth between those units (Max Bandwidth), or by reaching the maximum throughput of issuing memory instructions (Mem Pipes Busy). Detailed chart of the memory units. Detailed tables with data for each memory unit.

Memory Throughput [Gbyte/s]	51.11	Mem Busy [%]	6.79
L1/TEX Hit Rate [%]	98.64	Max Bandwidth [%]	11.72
L2 Hit Rate [%]	82.17	Mem Pipes Busy [%]	12.33
Local Memory Spilling Requests	0	Shared Memory Spilling Requests	0
Local Memory Spilling Request Overhead [%]	0	Shared Memory Spilling Request Overhead [%]	0
L2 Sector Promotion Misses [%]	15.17		

OK **Local Memory Usage Est. Speedup: 20.23%** This workload accesses 89 bytes per instruction, accounting for 7.43% of all sectors requested in L1/TEX. Local memory refers to a memory space that is private to each thread. Although it is logically distinct, local memory resides in global memory. This can lead to higher memory latency compared to the usage of registers or shared memory. Consider reducing the amount of local memory used.
7.43% of all local loads and 33.15% of all local stores spill into the L1 cache, accounting for 1.25% of all sectors requested in L2.
One common reason for the use of local memory is arrays being too large to fit into registers or not being indexed using compile-time constants (see the 'Local Memory' section of the CUDA Programming Guide). For more advice on how to reduce the usage of local memory refer to the Documentation.

Performance Indicators

OK **L1TEX Global Load Access Pattern Est. Speedup: 8.61%** The memory access pattern for global loads from L1TEX might not be optimal. On average, only 11.5 of the 32 bytes transmitted per sector are utilized by each thread. This could possibly be caused by a stride between threads. Check the Address Counters section for uncoalesced global loads.

Performance Indicators

OK **L1TEX Global Store Access Pattern Est. Speedup: 4.26%** The memory access pattern for global stores to L1TEX might not be optimal. On average, only 21.9 of the 32 bytes transmitted per sector are utilized by each thread. This could possibly be caused by a stride between threads. Check the Address Counters section for uncoalesced global stores.

Performance Indicators

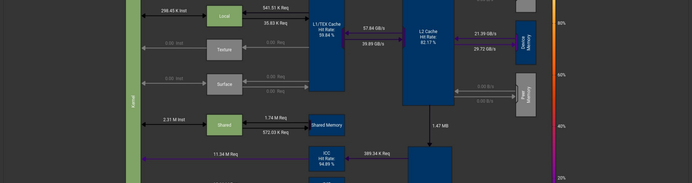
OK **L1TEX Local Load Access Pattern Est. Speedup: 6.26%** The memory access pattern for local loads from L1TEX might not be optimal. On average, only 17.1 of the 32 bytes transmitted per sector are utilized by each thread. This could possibly be caused by a stride between threads. Check the Address Counters section for uncoalesced local loads.

Performance Indicators

OK **L1TEX Local Store Access Pattern Est. Speedup: 13.01%** The memory access pattern for local stores to L1TEX might not be optimal. On average, only 1.1 of the 32 bytes transmitted per sector are utilized by each thread. This could possibly be caused by a stride between threads. Check the Address Counters section for uncoalesced local stores.

Performance Indicators

Memory Chart



Scheduler Statistics

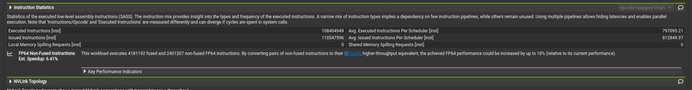
Summary of the activity of the schedulers issuing instructions. Each scheduler maintains a pool of warps that it can issue instructions for. The upper bound of warps in the pool (Theoretical Warps) is limited by the launch configuration. On every cycle each scheduler checks the state of the allocated warps in the pool (Active Warps). Active warps that are not stalled (Eligible Warps) are ready to issue their next instruction. From the set of eligible warps the scheduler selects a single warp from which to issue one or more instructions (Issued Warp). On cycles with no eligible warps, the issue slot is skipped and no instruction is issued. Having many skipped issue slots indicates poor latency hiding.

Active Warps Per Scheduler [warp]	2.89	No Eligible [%]	91.01
Eligible Warps Per Scheduler [warp]	0.10	One or More Eligible [%]	8.99
Issued Warp Per Scheduler	0.09		

OK **Issue Slot Utilization Est. Local Speedup: 64.87%** Every scheduler is capable of issuing one instruction per cycle, but for this workload each scheduler only issues an instruction every 11.1 cycles. This might leave hardware resources underutilized and may lead to less optimal performance. Out of the maximum of 8 warps per scheduler, this workload allocates an average of 2.89 active warps per scheduler, but only an average of 0.10 warps were eligible per cycle. Eligible warps are the subset of active warps that are ready to issue their next instruction. Every cycle with no eligible warp results in no instruction being issued and the issue slot remains unused. To increase the number of eligible warps, reduce the time the active warps are stalled by inspecting the top stall reasons on the Warp State Statistics and Source Counters sections.

Performance Indicators

Warps Per Scheduler



Warp State Statistics

Analysis of the states in which all warps spent cycles during the kernel execution. The warp states describe a warp's readiness or inability to issue its next instruction. The warp cycles per instruction define the latency between two consecutive instructions. The higher the value, the more warp parallelism is required to hide this latency. For each warp state, the chart shows the average number of cycles spent in that state per issued instruction. Stalls are not always impacting the overall performance nor are they completely avoidable. Only focus on stall reasons if the schedulers fail to issue every cycle. When executing a kernel with mixed library and user code, these metrics show the combined

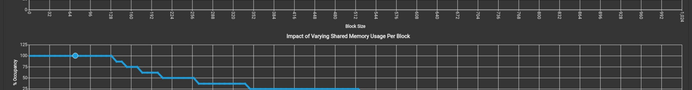
Warp Cycles Per Issued Instruction [cycle]	32.20	Avg. Active Threads Per Warp	27.34
Warp Cycles Per Executed Instruction [cycle]	32.84	Avg. Not Predicated Off Threads Per Warp	24.16

OK **Long Scoreboard Est. Speedup: 51.81%** On average, each warp of this workload spends 16.7 cycles being stalled waiting for a scoreboard dependency on a L1TEX (local, global, surface, texture) operation. Find the instruction producing the data being waited upon to identify the culprit. To reduce the number of cycles waiting on L1TEX data accesses verify the memory access patterns are optimal for the target architecture, attempt to increase cache hit rates by increasing data locality (coalescing), or by changing the cache configuration. Consider moving frequently used data to shared memory. This stall type represents about 51.8% of the total average of 32.2 cycles between issuing two instructions.

Performance Indicators

Warp Stall Check the Warp Stall Sampling (All Samples) table for the top stall locations in your source based on sampling data. The Profiling Guide provides more details on each stall reason.

Warp State (All Cycles)



Instruction Statistics

Global Category Chart

Statistics of the executed low level assembly instructions (SASS). The instruction mix provides insight into the types and frequency of the executed instructions. A narrow mix of instruction types implies a dependency on few instruction pipelines, while others remain unused. Using multiple pipelines allows hiding latencies and enables parallel execution. Note that 'Instructions/Opcode' and 'Executed Instructions' are measured differently and can diverge if cycles are spent in system calls.

Executed Instructions [inst]	108404949	Avg. Executed Instructions Per Scheduler [inst]	797995.21
Issued Instructions [inst]	110567896	Avg. Issued Instructions Per Scheduler [inst]	812849.97
Issued Warps Per Scheduler [inst]	0	Shared Memory Spilling Requests [inst]	0

OK **FP64 Non-Fused Instructions Est. Speedup: 5.41%** This workload executes 4181192 fused and 2401207 non-fused FP64 instructions. By converting pairs of non-fused instructions to their HFMA higher-throughput equivalent, the achieved FP64 performance could be increased by up to 18% (relative to its current performance).

Performance Indicators

NVLink Topology

NVLink Topology diagram shows logical NVLink connections with transmit/receive throughput.

NVLink Tables

Detailed tables with properties for each NVLink.

NUMA Affinity

Non-uniform memory access (NUMA) affinities based on compute and memory distances for all GPUs.

Launch Statistics

Summary of the configuration used to launch the kernel. The launch configuration defines the size of the kernel grid, the division of the grid into blocks, and the GPU resources needed to execute the kernel. Choosing an efficient launch configuration maximizes device utilization.

Grid Size	98	Function Cache Configuration	CachePreferNone
Registers Per Thread [register/thread]	64	Static Shared Memory Per Block [Kbyte/block]	4.61
Block Size	128	Dynamic Shared Memory Per Block [Byte/block]	0
Threads [thread]	12544	Driver Shared Memory Per Block [Byte/block]	0
Waves Per SM	0.36	Shared Memory Configuration Size [Kbyte]	65.54
Uses Green Context	0	Stack Size	1024
# SMs [SM]	34	# TPCs	17
Threads Per Block	128		

Occupancy

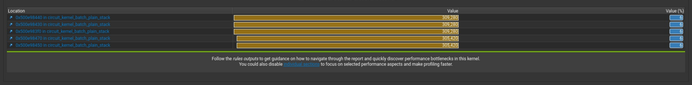
Occupancy is the ratio of the number of active warps per multiprocessor to the maximum number of possible active warps. Another way to view occupancy is the percentage of the hardware's ability to process warps that is actively in use. Higher occupancy does not always result in higher performance, however, low occupancy always reduces the ability to hide latencies, resulting in overall performance degradation. Large discrepancies between the theoretical and the achieved occupancy during execution typically indicates highly imbalanced workloads.

Theoretical Occupancy [%]	100	Block Limit Registers [block]	8
Theoretical Active Warps per SM [warp]	32	Block Limit Shared Mem [block]	13
Achieved Occupancy [%]	36.11	Block Limit Warps [block]	8
Achieved Active Warps Per SM [warp]	11.58	Block Limit SM [block]	16

OK **Achieved Occupancy Est. Speedup: 63.89%** The difference between calculated theoretical (100.0%) and measured achieved occupancy (36.1%) can be the result of warp scheduling overheads or workload imbalances during the kernel execution. Load imbalances can occur between warps within a block as well as across blocks of the same kernel. See the Profiling Guide for more details on optimizing occupancy.

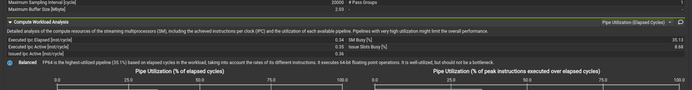
Performance Indicators

Impact of Varying Register Count Per Thread



Impact of Varying Block Size

Impact of Varying Shared Memory Usage Per Block



GPU and Memory Workload Distribution

Analysis of workload distribution in active cycles of SM, SMP, SMSP, L1 & L2 caches, and DRAM

Average SM Active Cycles [cycle]	9075575.32	Average L1 Active Cycles [cycle]	9075575.32
Average L2 Active Cycles [cycle]	6994084.56	Average SMSP Active Cycles [cycle]	9042694.85
Average DRAM Active Cycles [cycle]	5094425.50	Total SM Elapsed Cycles [cycle]	318450674
Total L1 Elapsed Cycles [cycle]	318450674	Total L2 Elapsed Cycles [cycle]	209438032
Total SMSP Elapsed Cycles [cycle]	1275790696	Total DRAM Elapsed Cycles [cycle]	349911296

Workload Distribution

	Average	Min	Max	Sum
SM Active Cycles	9075575.32	8037869	9397244	308549951
SMSP Active Cycles	9042694.85	7983506	9379944	1232775500
L1 Active Cycles	9075575.32	8037869	9397244	308549951
L2 Active Cycles	6994084.56	6887232	7235025	237819706
DRAM Active Cycles	5094425.50	4904772	5184402	46355364

Source Counters

Source metrics, including branch efficiency and sampled warp stall reasons. Warp Stall Sampling metrics are periodically sampled over the kernel runtime. They indicate when warps were stalled and couldn't be scheduled. See the documentation for a description of all stall reasons. Only focus on stalls if the schedulers fail to issue every cycle

Branch Instructions [inst]	15020008	Branch Efficiency [%]	98.28
Branch Instruction Ratio [%]	0.14	Avg. Divergent Branches [branches]	1152.68

OK **Uncoalesced Global Accesses Est. Speedup: 14.43%** This workload has uncoalesced global accesses resulting in a total of 5309893 excessive sectors (18% of the total 28758471 sectors). Check the L2 Theoretical Sectors Global Excessive table for the primary source locations. The CUDA Programming Guide has additional information on reducing uncoalesced device memory accesses.

Performance Indicators

L2 Theoretical Sectors Global Excessive

Location	Value	Value (%)
0x5000f0440 in circuit_kernel_batch_plain_stack	309,248	6
0x5000f0450 in circuit_kernel_batch_plain_stack	309,248	6
0x5000f0470 in circuit_kernel_batch_plain_stack	309,248	6
0x5000f0410 in circuit_kernel_batch_plain_stack	309,248	6
0x5000f0400 in circuit_kernel_batch_plain_stack	309,248	6

Follow the runbook to get guidance on how to navigate the report and quickly discover performance bottlenecks in this kernel.
You could also disable runbook guidance to focus on selected performance aspects and make profiling faster.